



Student Name: Valli Karthik

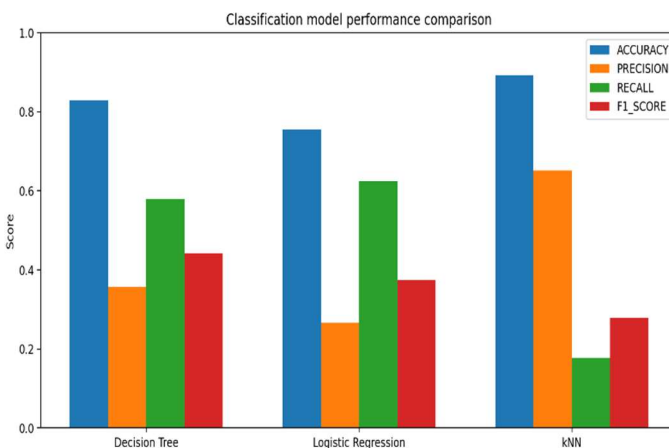
Course Code: DSA0402

Register number :192324258

Course Name: FUNDAMENTALS OF DATA SCIENCE

Course Faculty: Dr M K Sundaram

Project Title: Bank Customer Subscription Prediction and Customer Segmentation Using Data Science and Machine Learning



```
===== BANK CUSTOMER ANALYTICS MENU =====
1. Show dataset summary
2. Show subscription distribution
3. Show model comparison
4. Show cluster profiles
5. Show correlation matrix
6. Predict subscription for a new customer
0. Exit
Enter choice: 6

--- New Customer Subscription Prediction ---
Using best model: Decision Tree
Enter customer details. Do not enter 'duration' because it is excluded
from prediction to avoid target leakage.

Age: 35
Job: management
Marital (married/single/divorced): married
Education (primary/secondary/tertiary/unknown): tertiary
Default (yes/no): no
Balance: 2500
Housing loan (yes/no): yes
Personal loan (yes/no): no
Contact (cellular/telephone/unknown): cellular
Last contact day (1-31): 15
Last contact month (jan-dec): may
Number of contacts in this campaign: 2
Days since previous campaign contact (-1 if never contacted): -1
Number of previous contacts: 0
Previous outcome (failure/other/success/unknown): unknown

=====
PREDICTION RESULT
=====
Customer is predicted to: NOT SUBSCRIBED
Probability of subscription: 35.34%
Best model used: Decision Tree
```

Project Description:

This project is developed to predict whether a bank customer is likely to subscribe to a term deposit based on their demographic, financial, and previous campaign information. The system performs data preprocessing, exploratory data analysis, and statistical analysis to understand customer behaviour. Three machine learning algorithms kNN, Decision Tree, and Logistic Regression—are used to make predictions and compare their performance. In addition, K-means clustering is used to group customers with similar characteristics. The results are used to identify potential customer segments and provide data-driven marketing recommendations to improve campaign effectiveness.

V.Karthik....
Student Signature

Guide Signature